

3. (a) A group of students investigated the effect of pollution on species diversity in a stream. They sampled an unpolluted stream and a polluted stream and recorded the number of individuals of different species present in each.

(i) Identify the dependent and independent variables in their study. [2]

Dependent variable =

Independent variable =

(ii) Identify **one** possible source of error when investigating the biodiversity of animals. [1]

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(iii) The results for the unpolluted stream are given in the table below.

Species. Common name with description	<i>n</i>	(<i>n</i> -1)	<i>n</i> (<i>n</i> -1)
Water boatman. Surface living, fast swimming.	4	3	12
Mayfly nymph. Bottom dwelling, fast swimming.	30	29	870
Freshwater shrimp. Bottom dwelling, fast moving.	70	69	4 830
Water louse. Bottom dwelling, fast moving.	34	33	1 122
Bloodworm. Bottom dwelling in mud. Slow moving.	10	9	90
Sludgeworm. Bottom dwelling in mud. Slow moving.	2	1	2
	N =		Σ <i>n</i> (<i>n</i> -1) =
	N (N – 1) =		

N = total number of individual animals of all species.
n = number of individuals per species of each species.
Σ = sum of



Use the formula given below to calculate the Diversity Index for the unpolluted stream.
Give your answer to 2 dp. [3]

Diversity Index = $1 - \frac{\sum n(n - 1)}{N(N - 1)}$

(iv) The diversity index of the polluted stream was 0.1. What can you conclude from this figure? [1]

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(v) One student calculated the diversity index in one section of a stream to be 10.24.
Explain what can be concluded from this result. [2]

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- (b) The banded snail (*Cepaea nemoralis*) is found in a variety of habitats, such as grassland, heathland, sand dunes and the base of hedges. The shell colour can vary:



- yellow (which appears green with the animal inside)
- brown
- pink
- white

The shell can be unbanded or have up to five bands.

- (i) These morphologically different snails are all the same species. Explain why *Cepaea nemoralis* is an example of 'genetic polymorphism'. [2]

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The students made the following observations over a period of time:

- the main predator of the banded snail is the thrush
- fewer yellow shelled snails are eaten in summer than in winter but this is the opposite for banded snails
- in habitats such as tangled, dense hedges, dark banded snails are not eaten but large numbers of yellow snails are
- very few yellow snails are eaten from grassland or sand dunes but banded are

- (ii) Explain how genetic polymorphism is maintained in banded snail populations. [3]

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